GIT FUNDAMENTALS

SOURCE CONTROL

IN THIS PRESENTATION

- Source Control Management
- Git Fundamentals
- Git Commands

SOURCE CONTROL MANAGEMENT(SCM)

- SCM is a process that manages a collection of source code and changes
- Allows us to maintain multiple versions of code and code history
  - We can go back to any previous version of code if we break something
- Enables developers to work in parallel, even across the globe
- Synchronizes the code and identifies differences between versions
- Capable of handling both small and large projects
- Helps prevent code loss

SOURCE CONTROL MANAGEMENT(SCM)

- There are 2 types of version control systems:
  - Centralized version control – operates on a client-server relationship where the server is the repository and developers push their source code to this remote repository
  - Distributed version control – each user has a local copy of the repository; users can make commits, branches, and merges all on their local machine

GIT FUNDAMENTALS

- Git is one of the most popular source control tools and we will be using it for this course
- Git is a distributed version control system (local) while GitHub is a centralized VCS (remote)
- A Git project has 3 main sections:
  - The working tree/working directory – a single checkout of one version of the project on your local machine; set of all files and folders a developer can add, edit, rename, and delete
  - The staging area – a file that stores information about what will go into your next commit; technically called the 'index'
  - The Git directory – where Git stores the metadata and object database for your project (the most important part of Git!) This is what is copied when you clone a repository

GIT COMMANDS

- Git Cheat Sheet

GIT DEMO

- Let's initialize a repository and push to a remote repository

CHALLENGE

- Open GitBash and clone the repository I share
- Clone the remote repository
- Check the repo status
- Make a change
  - Open team.txt and add your name on a new line under 'Team Members:'
  - Save the file
- Stage your changes
- Commit your changes
- Pull from GitHub
- Push to GitHub